Jewoo Suh

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Research Interests: Knowledge Representation & Reasoning, Evolutionary Algorithms, Mathematics & Computer Science

Education

University of Oxford, *Christ Church College* 2021.10 – 2025.07
Integrated Master of Mathematics and Computer Science (MMathCompSci)
Thesis: Designing the VG8: An FPGA-Based 8-bit Game Console
Classification: Second Class Honours, Division I (2:1) in Parts A & B; Merit in Part C

Work Experience

LIG Defense & Aerospace (LIG D&A) *Pangyo, Republic of Korea* 2025.07 – Present
Research Engineer @ Technology Innovation Division, AI R&D Team
Serving as Technical Research Personnel (mandatory military service) from 2025.07.

- Develop and maintain the Next Software Framework (NSFW), a multi-OS internal framework deployed across various defence systems for core mission-critical functionality.
- Collaborate with Palantir Technologies on AI-based command-and-control and multi-agent coordination for autonomous maritime and multi-domain platforms.
- Participating in development of a company-wide internal AI Toolkit to standardise AI integration across teams.

Xsolla Korea *Seolleung, Seoul, Republic of Korea* 2023.07 – 2024.12
Intern → Process Automation Manager

- Designed and deployed core AWS infrastructure for Xsolla's APAC branch.
- Automated and maintained internal Salesforce database organisation, reducing manual overhead across the regional office.

Worked full-time during university summer breaks and part-time during term-time.

Publications

[1] J. Suh et al., "Iterated Self-Evaluation Impossibility for Large Language Models," *in preparation*, IEEE Access, 2026.

[2] J. Suh et al., "Byzantine Monitors in AI Control: A Threshold Hierarchy for Budgeted Oversight," *in preparation*, Journal of Artificial Intelligence Research (JAIR), 2026.

[3] J. Suh, "Truthful Task and Resource Allocation in Leader–Follower Agent Organisations," *in preparation*, Autonomous Agents and Multi-Agent Systems (JAAMAS), 2026.

[4] J. Suh, "Register-Invariant Jamo Coding for Short Korean Text," *in preparation*, ICTC 2026.

[5] W.-S. Kim et al. (incl. J. Suh), "[Digital Twin-Verified Design and Development of a USV Swarm Command and Control System]," *in preparation*, JKIICE, 2026.

[6] J. Suh, "Designing the VG8: An FPGA-Based 8-bit Game Console for Comparative Analysis of 1980s Personal Computers and Game Consoles," MMathCompSci Thesis, University of Oxford, 2025.

Projects

Cross-Ocean Passive SONAR Classification: Evolved interpretable signal frontends for vessel classification. *ongoing*

Evolved Swarm Behaviour: Emergent leaderless drone coordination from a single evolved shared controller. *ongoing*

Knowledge Classification Project: Cross-field knowledge formalisation via formal publication language. *ongoing*

Language Evolution Simulator: Emergent grammar and lexicon via genetic algorithms. *ongoing*

Etymology Tree Visualisation: Interactive etymology explorer. *completed*

Voynich Decode Project: Statistical structure analysis of the Voynich Manuscript. *closed*

For further details on publications and projects, visit jewoo-suh.github.io.

Honours & Awards

Minister's Award, Ministry of Science and ICT 2025
2025 Republic of Korea SW Technology Grand Prize, IT Architecture Category

Skills

AI Tools: Claude Code, Gemini, GPT Codex
Programming: Python, Verilog, C, Scala, Unity/C#, Java, Haskell, MATLAB, LaTeX, HLSL
Languages: English (Fluent), Korean (Fluent), Chinese (Conversational), Japanese (Conversational)